Section II: Distribution Fitting and Normality Testing of Bitcoin and Ethereum Returns

Overview

A generalized normal distribution fits BTC/USD one-minute returns best, and the Shapiro-Wilk test rejects normality for ETH five-minute returns in all 200 random samples tested. Both analyses use log returns of closing prices, $r(t) = \ln(P(t)/P(t-1))$, computed in R.

Series	Interval	Period analysed	Returns (n)	Exactly-zero returns
BTC/USD	1 minute	2012-01-01 to 2026-09-15	7,735,678	24.7%
ETH	5 minutes	2017-08-17 to 2026-09-06	950,940	1.1%

The ETH file begins on 2017-08-17 and ends on 2026-09-06, so that is the coverage analysed, not the requested 2015-08-07 to 2026-09-15.

Part 1 fits eight candidate distributions by maximum likelihood and ranks them by AIC and BIC, with the Kolmogorov-Smirnov (KS) statistic as a check. Part 2 applies the Shapiro-Wilk test, supported by the Jarque-Bera test.

Part 1: Best-fitting distribution for BTC/USD 1-minute returns

The generalized normal distribution fits best, ahead of Johnson SU by 546 AIC points, with an estimated shape parameter of 0.54. All eight distributions were fitted by maximum likelihood to a random sample of 150,000 of the 5,825,336 non-zero returns.

Distribution	Parameters (k)	Log-likelihood	AIC	BIC	ΔAIC	KS statistic
Generalized normal	3	812,790	-1,625,574	-1,625,545	0	0.0097
Johnson SU	4	812,518	-1,625,028	-1,624,988	546	0.0150
Student-t	3	811,187	-1,622,369	-1,622,339	3,206	0.0211
Cauchy	2	809,401	-1,618,797	-1,618,777	6,777	0.0171

Distribution	Parameters (k)	Log-likelihood	AIC	BIC	ΔAIC	KS statistic
Laplace	2	798,510	-1,597,017	-1,596,997	28,558	0.0570
Hyperbolic secant	2	788,389	-1,576,774	-1,576,754	48,800	0.0796
Logistic	2	780,159	-1,560,314	-1,560,294	65,261	0.0950
Normal	2	726,629	-1,453,255	-1,453,235	172,320	0.1794

Best-fit generalized normal parameters: location 1.6×10^{-7}, scale 1.9×10^{-4}, shape 0.537.

Interpretation. The generalized normal ranks first on AIC, BIC and the KS distance. A shape of 2 is the Normal and 1 is the Laplace, so a shape of 0.54 means a sharper peak at zero and heavier tails than either. The Normal ranks last ($\Delta\text{AIC} = 172,320$), and its Q-Q curve is strongly S-shaped, the typical signature of fat tails.

Two cautions apply. The KS test rejects every candidate ($p \approx 0$), which is expected with millions of discrete observations, so "best" means closest among those tested. The density plot also suggests the generalized normal falls below the observed frequencies in the far tails, where Johnson SU and Student-t follow the data more closely.

Part 2: Shapiro-Wilk normality test for ETH 5-minute returns

Normality is rejected: all 200 random samples of 5,000 returns gave $p < 0.05$, with a mean W of 0.779. The null hypothesis is that returns are normally distributed. Shapiro-Wilk accepts at most 5,000 observations, so it was applied to random samples of that size and repeated 200 times.

Statistic	Value
Returns analysed	950,940 (2017-08-17 to 2026-09-06)
Shapiro-Wilk, one random sample of 5,000	$W = 0.855$, $p = 1.15 \times 10^{-55}$
Mean W over 200 random samples	0.779

Statistic	Value
Median p-value over 200 random samples	9.4×10^{-63}
Samples rejecting normality at 5%	200 of 200
Jarque-Bera (all returns)	3.45×10^{8}, $p < 0.001$
Skewness	-0.156
Excess kurtosis	93.4

Interpretation. Normality is rejected at any usual significance level. Every repetition rejected, so the result does not depend on one lucky draw; W for a single sample varies between draws, and the 200-sample mean is the stable summary. The Jarque-Bera test agrees.

The returns are nearly symmetric (skewness -0.16) but extremely fat-tailed (excess kurtosis 93.4; a Normal distribution has 0). The histogram shows a sharper peak than the fitted Normal curve, and the Q-Q plot bends away from the line in both tails: extreme moves occur far more often than a Normal distribution allows.

Limitations and notes

- **ETH coverage.** The file runs from 2017-08-17 to 2026-09-06, not the requested 2015-08-07 to 2026-09-15.
- **Zero returns in BTC.** 24.7% of one-minute returns are exactly zero (price unchanged). A continuous density cannot describe that point mass, and including it made every fit collapse onto the spike, so distributions were fitted to the 5,825,336 non-zero returns.
- **Subsampling.** To keep fitting tractable, each distribution was fitted to a random sample of 150,000 non-zero BTC returns (fixed random seed). Log-likelihood, AIC, BIC and KS refer to that sample.
- **Extreme values.** BTC contains single-minute returns of about ± 62, which look like data errors. They were not removed, and they drive the excess kurtosis of 7,099 in the full series.
- **Missing minutes.** Returns are computed between consecutive available rows; missing intervals were not filled.
- **Software.** R Studio